Alex Cason

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Physics-is-awesome
YouTube: Eudaimonia_society

Education

Expected May 2030 **Bachelor of Science in Physics and Mathematics**, *University of Tennessee*, Knoxville, TN

Technical Skills

Coding **Experienced:** FORTRAN, Python, LaTeX **Comfortable:** Julia, Bash, Lisp, OpenMP **Basic:** C++, C

Software Flash-X, OpenMHD, Gkeyll, PlasmaPy, SPIS, OpenFoam, Pluto, Einstein's Toolkit, Linux

Research & Technical Experience

Aug 2024–Present **Assistant Researcher**, *University of Tennessee*, Knoxville, TN, Dr. Mezzacappa

- Research gravitational waves resulting from core-collapse supernovae.
- Develop code to process core-collapse supernova simulation data (generated via FLASH-X) to extract and analyze gravitational wave data.
- Now working towards developing tools to observe core-collapse supernova wave memory in next generation detectors.

Aug 2026–Present **Nuclear Electric Propulsion Thruster Design Subteam Lead**, *SSTA*

- Schedule and lead weekly technical meetings.
- Instruct students in fundamentals of plasma physics, numerical methods, and coding.
- Plan project goals and technical direction utilizing numerical software expertise.

Nov 2024–Present **Contributor**, *PlasmaPy*

- Enhance computational tools and simulation frameworks for plasma physics research.
- Engage in active scientific discussion and comprehensive code reviews.
- Develop and refine educational tools for the plasma physics community.

May 2026–Present **Content Creator**, *YouTube*

- Develop advanced educational videos on physics topics including General Relativity, metriplectic dynamics, and Hamiltonian Dynamics.
- Cover complex derivations such as Navier-Stokes from Boltzmann, and Hamiltonian dynamics.
- Demonstrate technical projects including Discontinuous Galerklin, BSSN codes, and advanced symplectic integrators and geometric numerical methods.

Leadership & Extracurricular Experience

Aug 2024–May 2026 **Founder**, *Scientific Visionaries of America*

- Organized the inaugural Wilson County Science Fair.
- Taught advanced scientific subjects and prepared members for rigorous competitions (JSHS, Science Olympiad, Science Bowl, Science and Engineering Fair).

Oct 2024–May 2026 **Chapter Chief & Fall Fellowship Director**, *Order of the Arrow*

- Managed chapter operations, including a team of officers (Secretary, Treasurer, Vice Chief).
- Planned and executed large-scale events, campouts, and community service projects.
- Led over 30 members in the execution of the Fall Fellowship for over 211 attendees.

Aug 2024–May 2026 **S1 Officer**, *JROTC*

Managed administrative paperwork, tracked attendance and service hours, and planned physical training sessions.

Personal Projects & Models

Physics/Math	Hamiltonian Framework on ICF Implosions Rocket Equation, Relativistic BSSN, Numerical Methods for 3D Compressed Plasma using Lattice Boltzmann, Relativistic Rocket Equation (Hamiltonian), WHCKL Celestial Mechanics Simulation.
Computational	Simulated Annealing models, PPO AI to play Asteroid game, DG-CCSN, Planck Force Simulation using Einstein's toolkit.
MHD/Plasma	MHD FORTRAN model, Hamiltonian MHD, MHD Rocket Simulation, Metriplectic Plasma Simulation, Variational MHD Simulation.

Awards

2026	Vigil Honor Award (Order of the Arrow)
2025	Eagle Scout, Naval Science Award, 2nd Place Regional Oratorical
2024	2nd Degree Black Belt (Karate)

Community Service

Providence Church	Assisted teaching Sunday School, led youth sermons, served communion, and delivered food to the elderly and homeless.
Scouting/OA	Staffed multiple regional camps (Winter camp, Klondike), cleaned parks/reservations, and assisted in Eagle Scout workdays.